GOVIND UNNIKRISHNAN

Physics BS-MS Graduate

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EDUCATION

Integrated BS-MS (Physics Major, Data Science Minor) 2021 – 2026
Indian Institute of Science Education and Research, Thiruvananthapuram
CGPA: 8.42

RESEARCH EXPERIENCE

Low Temperature Adiabatic Demagnetization Refrigeration (ADR) Studies, *Dr. Deepshikha Jaiswal-Nagar*
Synthesized ADR materials via solid-state reactions and investigated their magnetic and thermodynamic properties. Performed adiabatic demagnetization refrigeration measurements on a candidate material, achieving temperatures down to 165 mK with a holding time exceeding 12 hours, demonstrating its suitability for magnetic refrigeration. Identified two field-induced critical points at millikelvin temperatures that enhanced entropy accumulation and contributed to the ultralow temperature performance. These features were confirmed through heat capacity measurements down to 100 mK, and H–T phase diagrams were constructed to map the critical behaviour and identify the quantum critical end point.

Crystal Electric Field Driven Effective Spin-1/2 Physics in a Rare-Earth Hyperkagome Lattice, *Dr. Deepshikha Jaiswal-Nagar*
Investigated effective spin-1/2 behaviour arising from crystal electric field (CEF) effects in the rare-earth hyperkagome lattice $Li_3Nd_3Te_2O_{12}$. Verified the stabilization of an effective spin-1/2 ground state through magnetization, heat capacity, and adiabatic demagnetization refrigeration (ADR) measurements. Analysed the CEF splitting and its role in governing the low-temperature thermodynamic response. Analyzed inelastic neutron scattering (INS) data to investigate the detailed crystal-field scheme and magnetic excitations of the rare-earth system.

High Entropy Alloys for Room Temperature Refrigeration Applications, *Dr. Deepshikha Jaiswal-Nagar*
Explored intermetallic compounds in varied stoichiometries to engineer and design rare-earth-free HEAs. Simulated and identified Tc using SPR-KKR (DFT) and Monte Carlo methods. Performed synthesis using arc melting and quartz sealing units, followed by structural and magnetic characterization.

Reinforcement Learning for Timetable Optimization, *Dr. Saptarshi Bej*
Developed a reinforcement learning algorithm to minimize timetable clashes, achieving a zero-clash schedule and extending the model for multi-slot subject optimization. Implemented parallelized optimization strategies and constraint handling via Lagrange multiplier-based walkers to efficiently explore the solution space and identify global minima.

INTERNSHIPS AND OTHER PROJECTS

Semester Project: Predicting Material Usability Using Machine Learning 01/2025 – 04/2025
Supervisor: Dr. Chandrakala Meena
- Developed machine learning models to predict material suitability for engineering applications using mechanical-property datasets.
- Trained and evaluated Logistic Regression, Random Forest, XGBoost, SVM, and ANN models on a dataset of 1552 materials; selected the optimal model based on accuracy and F1-score.

Internship Project: Low-Temperature Transport Measurements Using a Closed-Cycle Cryostat 10/2024 – 12/2024
- Manually operated a two-stage pulse-tube closed-cycle helium cryostat, including helium precooling, flow regulation, and VTI temperature control.
- Performed temperature- and magnetic-field-dependent 2-/4-probe transport measurements and developed MATLAB automation for instrument control and data acquisition.

Control Systems Project 09/2023 – 12/2023
- Designed and built a self-balancing robot and reaction wheel system using PID control.
- Programmed microcontrollers for real-time sensor integration and closed-loop control using Python and embedded systems

Quantum Materials Research, *Supervisor: Dr. Deepshikha Jaiswal Nagar* 04/2023 – 10/2023
- Synthesized $GdCrTiO_5$ (GCTO), and $Ba_6Yb_2Ti_4O_{17}$ (BYTO) for low-temperature magnetic studies.
- Investigated the magnetic ground state and spin-glass behavior of $ZnNiTiO_4$ (ZNTO) spinel.
- Assisted with the design and implementation of an adiabatic demagnetization refrigeration (ADR) puck for sub-kelvin measurements.

Solid State Physics Reading Project, *Supervisor: Prof. Ramesh Chandra Nath* 11/2022 – 01/2023

- Studied foundational concepts in solid-state physics through standard texts, including introduction to Solid State Physics (Kittel) and Solid State Physics (Ashcroft & Mermin).

EXPERIMENTAL EXPERIENCE

Material Synthesis

Experienced in solid-state reactions, single-crystal synthesis, arc melting, and tube sealing techniques for alloy synthesis, with hands-on experience in sputtering techniques for thin-film growth.

Cryogenic instrumentation

- Transport measurements (Performed R-T (2,4 probe and van der Pauw methods), Hall effect, and thermal transport studies.)
- Magnetic and thermodynamic measurements (PPMS,MPMS)
- Ultralow temperature transport and specific heat measurements (using ^{3}He-^{4}He dilution refrigerator)

Experimental Characterization

Powder XRD, single-crystal XRD, Rietveld refinement; ESR, XPS, Raman spectroscopy

COMPUTATIONAL AND THEORETICAL EXPERTISE

- **Electronic Structure & Magnetism:** DFT calculations using SPR-KKR and Quantum ESPRESSO for geometry optimization, SCF, NSCF, band structure, DOS/PDOS. spin-polarized and magnetic property analysis, crystal-field modeling using PyCrystalField.
- **Finite-Temperature & Spin Modeling:** Computation of exchange interactions via the LKAG formalism and Monte Carlo simulations to study magnetic phase transitions and estimate Tc.
- **Numerical & Data Analysis:** Proficient in Python and MATLAB for numerical analysis, data fitting, thermodynamic modeling, machine learning, and reinforcement learning.
- **Visualization & Post-Processing:** Experience with VESTA, Origin, GNUPlot, Tableau, GaussView and FullProf.

ACADEMIC AND EXTRACURRICULAR ACTIVITIES

- Volunteered at the Global Science Fest, demonstrating an Arduino-based project or human electromagnetic signal analysis using EOG and EMG data.
- Developed a Python script for automated certificate generation during the FSP Symposium
- Attended Frontier symposium in Physics 2025 held at IISER TVM from 17th - 19th January 2025. Frontier Symposium in Physics 2024 held at IISER TVM from 19th -21st January 2024. UGC DAE conference 14th December 2025, Conference on Quantum Materials and Devices on 9th August 2024 etc.
- Learned violin professionally and active football player

RELEVANT COURSES

- **Physics Major:** Condensed Matter Physics I & II, Quantum Mechanics I & II, Classical Mechanics, Mathematical Methods, Electronics, Semiconductor Physics, Statistical Mechanics, Electrodynamics and Special Theory of Relativity, Atomic and Molecular Physics, Computational Techniques and Programming Languages.
- **Data Science Minor:** Machine Learning, Scientific Computing, Data Analysis and Visualization
- **Electives:** Modelling Materials, Experimental Physics, Electrochemical Energy Systems, Machine Learning for Physical Sciences, Statistical and Data Analysis Methods in Physical Sciences, Advanced Physics Laboratories (4 undergraduate and 3 core labs).

PUBLICATIONS

Unraveling the Field-Induced Magnetism in the Frustrated Garnet Gd3Te2Li3O12 with Enhanced Adiabatic Demagnetization Refrigeration

Dheeraj Ranaut*, Govind Unnikrishnan, Samraj M., Sharath Kumar C., Prashanta K. Mukharjee, Philipp Gegenwart and Deepshikha Jaiswal-Nagar (Manuscript under review (accepted for publication in “ACS Applied Electronic Materials”))

REFERENCES

Dr. Deepshikha Jaiswal-Nagar, *Associate Professor,* IISER Thiruvananthapuram
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